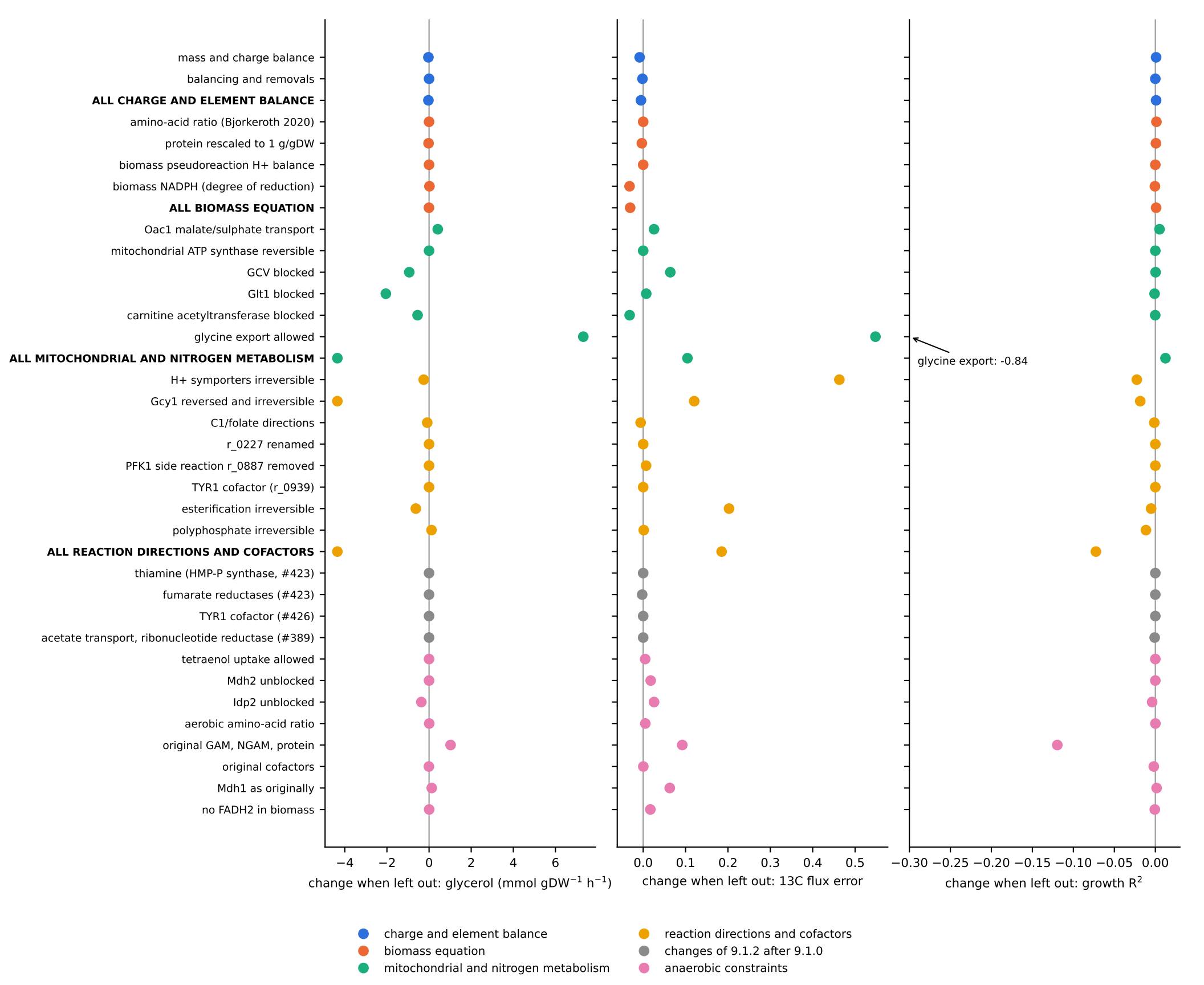
mass and charge balance
balancing and removals
ALL CHARGE AND ELEMENT BALANCE
amino-acid ratio (Bjorkeroth 2020)
protein rescaled to 1 g/gDW
biomass pseudoreaction H+ balance
biomass NADPH (degree of reduction)
ALL BIOMASS EQUATION
Oac1 malate/sulphate transport
mitochondrial ATP synthase reversible
GCV blocked
Glt1 blocked
carnitine acetyltransferase blocked
glycine export allowed
ALL MITOCHONDRIAL AND NITROGEN METABOLISM
H+ symporters irreversible
Gcy1 reversed and irreversible
C1/folate directions
r_0227 renamed
PFK1 side reaction r_0887 removed
TYR1 cofactor (r_0939)
esterification irreversible
polyphosphate irreversible
ALL REACTION DIRECTIONS AND COFACTORS
thiamine (HMP-P synthase, #423)
fumarate reductases (#423)
TYR1 cofactor (#426)
acetate transport, ribonucleotide reductase (#389)
tetraenol uptake allowed
Mdh2 unblocked
Idp2 unblocked
aerobic amino-acid ratio
original GAM, NGAM, protein
original cofactors
Mdh1 as originally
no FADH2 in biomass
glycine export: -0.84
−4 −2 0 2 4 6
change when left out: glycerol (mmol gDW^{-1} h^{-1})
0.0 0.1 0.2 0.3 0.4 0.5
change when left out: 13C flux error
−0.30 −0.25 −0.20 −0.15 −0.10 −0.05 0.00
change when left out: growth R^2
charge and element balance
biomass equation
mitochondrial and nitrogen metabolism
reaction directions and cofactors
changes of 9.1.2 after 9.1.0
anaerobic constraints